104: PYG MEDIUM (modified)

Final pH: 7.2
Final volume: 1000 ml

Trypticase peptone	5.00	g
Peptone	5.00	g
Yeast extract	10.00	g
Beef extract	5.00	g
Glucose	5.00	g
K_2HPO_4	2.00	g
Tween 80	1.00	ml
Salt solution	40.00	ml
Resazurin	1.00	mg
Vitamin K_1 solution	0.20	ml
Haemin solution	10.00	ml
L-Cysteine HCl x H_2O	0.50	g
Distilled water	950.00	ml

The vitamin K_1, haemin solution and the cysteine are added after the medium has been boiled and cooled under CO_2. Adjust pH to 7.2 using 8 N NaOH. Distribute under N_2 and autoclave.

For DSM 2008: Plus lactate (0.75%) plus putrescine (3 mg/l)

For DSM 2710: plus haemin plus VitK plus horse serum (5%)

For DSM 3133: Plus 3g/L cellubiose

For DSM 3353: With carbohydrates (see medium 110); anaerobic

For DSM 3985: Plus 0.5% maltose

For DSM 3988, DSM 3993, DSM 9848, DSM 18711, DSM 20482, DSM 20572, DSM 25383, DSM 26039, DSM 26228, DSM 43327: Plus haemine (5µg/ml) plus Vit K_1

For DSM 4000, DSM 10695, DSM 10702, DSM 20468, DSM 26657, DSM 43015, DSM 103158, DSM 103159, DSM 103345, DSM 103473, DSM 103940, DSM 114758, DSM 115111, DSM 117056, DSM 118263, DSM 119691, DSM 119692, DSM 119817, DSM 120170: plus haemine (5µg/ml) plus Vit K_1

For DSM 4901, DSM 9668, DSM 9670, DSM 13386, DSM 15643, DSM 15829, DSM 20362, DSM 20458, DSM 20614, DSM 20700, DSM 20706, DSM 21503, DSM 21784, DSM 25694, DSM 26980, DSM 43011, DSM 108211, DSM 108217, DSM 108236, DSM 108237, DSM 109769: Plus haemine (5µg/ml) plus Vit K_1

For DSM 6400: With 1 g/l of $NaHCO_3$; anaerobic

For DSM 6844, DSM 110989: plus haemin plus Vit K_1

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For DSM 9536: Plus disodium succinate 2.5g/l

For DSM 13220: (add 0.5 g/l Tween 80); anaerobic

For DSM 15923: Plus L-Arginine 1.0 g/L and sterile fetal bovine serum (100 µl/10 ml); anaerobic

For DSM 17217: Plus 10µl/ml putrescine (0.03%) plus 10 µl/ml sodium lactate (50%)

For DSM 17226: Plus fructose (2 g/L); anaerobic

For DSM 17234, DSM 17243: Plus bovine serum (10µl/ml)

For DSM 17476: Plus 10µl/ml bovine serum

For DSM 17678: Plus fetal bovine serum (1%)

For DSM 18011: Plus rumen fluid (200µl/10ml); anaerobic

For DSM 18206: Plus horse serum (5% vol/vol)

For DSM 18960, DSM 19857: Plus lactate (0.5 - 1%); anaerobic

For DSM 19346: plus vitamin solution (see medium 141)

For DSM 19450: Plus L-Arginin (1g/l)

For DSM 19757, DSM 19758: Anaerobic (90% N_2, 5% H_2, 5% CO_2)

For DSM 20361, DSM 103075, DSM 119816, DSM 120030, DSM 120038, DSM 121003, DSM 121043, DSM 121789, DSM 121931: plus hemin (5µg/ml) plus Vit K_1

For DSM 20476: Supplemented with arginine (2g/L); strictly anaerobic

For DSM 20732: Plus 5% fetal calf serum

For DSM 21469: Plus 5% horse serum

For DSM 21825: Plus 1% rumen fluid and volatile fatty acids (see medium 330)

For DSM 21845: Plus 1 % rumen fluid and volatile fatty acids (see medium 330); anaerobic

For DSM 21880: Anaerobic, 100% CO_2 atmosphere

For DSM 21887: Plus bovine serum (10µl/ml)

For DSM 22317: Plus Na-lactate (10 g/l)

For DSM 22474: Plus haemine plus Vit K 3; anaerobic

For DSM 22477: Plus 3 g/l arginine

For DSM 22478: Plus 1 g/l arginine

For DSM 22514: Plus arginine (1g/L)

For DSM 22521: Plus Na-lactate (0.75%)

For DSM 22533: Plus haemine plus Vit K_1 plus lactate (0.75%)

For DSM 22547: plus 5% bovine serum

For DSM 22619, DSM 22620: Adjust pH of final medium to 7.0.

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For DSM 22959: Plus 0.1% mucin

For DSM 23047, DSM 23056: Add 0.4% glucose, 0.1% cellobiose, 0.1% maltose, and 0.1% soluble starch; anaerobic

For DSM 23059: Plus 5% horse serum plus bicarbonate

For DSM 23669, DSM 25557, DSM 111893, DSM 111899: Supplement medium after autoclaving with 50.0 ml/l clarified rumen fluid (see medium 1310), 50.0 ml/l horse serum (Bio & Sell PFD.SE.0500), and 0.5 g/l dithiothreitol (DTT) added from filter-sterilized anoxic stock solutions.

For DSM 25311: Plus horse serum (5%) plus $NaHCO_3$ (0.1%)

For DSM 26143: Plus haemine (5µg/ml) plus Vit K_1; strictly anaerobic

For DSM 26962: Plus carbohydrate-mix (g/L): Glucose 4.0, cellobiose 1.0, maltose 1.0, starch (soluble) 1.0.; strictly anaerobic

For DSM 27982: Plus 15% rumen fluid

For DSM 100348: Supplemented with rumenfluid (30 ml/L)

For DSM 100440: Plus haemine (5µg/ml) plus Vit K_1; obligately anaerobic

For DSM 100705: Plus 0.05% mucin

For DSM 104750: Plus 0.25% Sodium lactate

For DSM 104782: Plus 0.5% fructose and 3% rumen fluid

For DSM 106041, DSM 110150, DSM 110159: plus haemine plus Vit K_1

For DSM 106790: Supplemented with sodium lactate (10g/L); strictly anaerobic

For DSM 108985: Anaerobic, N_2CO_2 plus H_2 (1 bar)

For DSM 109050: Plus haemine (50µg/ml) plus Vit K_1 plus Na-lactate (0.5%)

For DSM 109235: plus 0.5 g/L pectin

For DSM 111052: Plus haemine (5µg/ml) plus Vit K_1 plus Na-lactate (0.75%)

For DSM 114407: plus 0.1% mucin

For DSM 114537: dilute the medium 1:100

For DSM 114538: dilute the medium 1:10

For DSM 116972: Add 0.4% glucose, 0.1% cellobiose, 0.1% maltose, 0.1% soluble starch, and 2% rumen fluid

For DSM 118844: plus rumen fluid (200µl/10ml)

Salt solution (from medium 104)

$CaCl_2$ x 2 H_2O	0.25	g
$MgSO_4$ x 7 H_2O	0.50	g

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K_2HPO_4	1.00	g
KH_2PO_4	1.00	g
$NaHCO_3$	10.00	g
NaCl	2.00	g
Distilled water	1000.00	ml

Haemin solution (from medium 78)

Haemin	50.00	mg
NaOH (1 N)	1.00	ml
Distilled water	100.00	ml

Dissolve 50 mg haemin in 1 ml 1 N NaOH; make up to 100 ml with distilled water and filter sterilize. Store refrigerated.

Vitamin K_1 solution (from medium 78)

Vitamin K_1	0.10	ml
Ethanol (95 %)	20.00	ml

Dissolve 0.1 ml of vitamin K_1 in 20 ml 95% ethanol and filter sterilize. Store refrigerated in a brown bottle.

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